

Supply Chain Delay Prediction

Machine Learning Project Report

Prepared by: Muhammad Ayaan

A data-driven approach to optimizing logistics performance

Executive Summary

This project develops a machine learning model to predict shipment delays in a supply chain system. The model enables proactive decision-making, helping businesses reduce delays and improve customer satisfaction. The current solution achieves 68% accuracy with strong performance in identifying delayed shipments.

Problem Statement

Delivery delays negatively impact customer experience and operational efficiency. This project aims to classify whether an order will be delayed using historical logistics data.

Data & Features

The dataset includes operational and transactional features such as shipping mode, product category, region, and delivery timelines. Both categorical and numerical variables were used for modeling.

Methodology

The workflow includes data preprocessing, encoding categorical variables, handling missing values, balancing the dataset using SMOTE, and training a Random Forest classifier.

Model Performance

Metric	Score
Accuracy	68%
Precision	73%
Recall	72%
F1 Score	68%

Key Insights

The model demonstrates higher accuracy in predicting delayed shipments compared to on-time deliveries. This makes it useful for risk identification but highlights the need for improved balance.

Limitations

Performance is constrained by limited feature engineering and encoding techniques. Frequency encoding may not fully capture categorical relationships.

Recommendations

Future improvements include advanced encoding methods, feature engineering, and using boosting algorithms such as XGBoost for better predictive performance.

Conclusion

This project showcases the application of machine learning in supply chain analytics. With further enhancements, the model can become a valuable decision-support tool in logistics operations.